

Reading Tables, Charts, Graphs, and Photos

Curator's Question

How do tables, charts, graphs, and photos help curators tell a fuller story than maps alone?

MAP CURATOR

ENTER THE SCENE

More Than Just Maps

The Greenland explanation card is up — visitors finally understand why the flat map and globe look different. But look around the Great Map Room. It's not just maps in here.

There's a bar chart showing how many people live on each continent. A table listing the world's longest rivers. A photograph of a mountain range. A line graph tracking temperatures over the year.

Maps tell part of the story, but curators use all kinds of evidence to help visitors learn. Today you'll practice reading these other tools so the whole room makes sense together.

Self-Governance

Making good choices on your own.

When curators decide which chart, graph, or photo to study, they're governing their own thinking. They don't grab the first thing they see — they stop, think about the question, and pick the evidence that actually helps.

TODAY'S CONNECTION

Today you'll face different types of evidence — tables, charts, graphs, photos. Self-governance means you choose the right evidence for the question instead of just picking the first one you see.

Discovery Clue: What's in This Display?

INSERT IMAGE

Museum wall showing a variety of displays: a table of data, a bar chart, a line graph, a large photograph, and a map — all arranged as museum exhibits with labels

The Great Map Room has more than maps!

Maps show **WHERE**.

But what about...

HOW MANY?

HOW THINGS CHANGE?

WHAT IT REALLY LOOKS LIKE?

SOCRATIC QUESTIONS

- What kinds of evidence do you see in museums besides maps?
- Why might a museum need more than one type of display?

Discovery Reveal: Tables

TABLE

Organizes facts into rows and columns so you can find and compare information quickly.

Example: A table of the world's longest rivers — with river name, length, and continent.

Tables show WHAT FACTS

They answer questions like: How long is the Nile? Which continent has the Amazon? What are the top five longest rivers?

INSERT IMAGE

Simple, clean table showing 4-5 of the world's longest rivers with columns: River Name, Length (miles), Continent

SOCRATIC QUESTIONS

- *What makes a table different from just writing facts in a paragraph?*
- *When would rows and columns help you find an answer faster?*

Chat Frame: Table Reading

CHAT FRAME

"If you wanted to find which river is the longest, would you look at a map or a table? Why?"

THINK ABOUT IT

A **map** can show you where a river is, but can it tell you exactly how long it is in miles?

A **table** lists the exact length right next to the name — no measuring needed.

Discovery Reveal: Bar Charts

BAR CHART

Uses bars of different heights to compare amounts. Taller bar = more.

Example: Population of each continent — the tallest bar shows the most people.

Charts show HOW MUCH

They answer questions like: Which continent has the most people? How do Africa and Europe compare? Is South America bigger or smaller than North America in population?

INSERT IMAGE

Colorful bar chart showing population by continent — Asia's bar clearly tallest, Antarctica very small, clear labels and title

SOCRATIC QUESTIONS

- *Why is it easier to compare amounts with bars than with a list of numbers?*
- *What does the height of a bar tell you?*

Discovery Clue: The Line Goes Up and Down

A visitor walks up and asks:

“Has the temperature been going up over the year?”

Could a table answer this question?

A table could list the temperatures, but you'd have to read every number and figure out the pattern yourself.

What if there was a picture that SHOWED you the ups and downs?

INSERT IMAGE

Simple line graph showing average monthly temperature over 12 months — line rises from January, peaks in summer, falls again — clean labels on axes

Discovery Reveal: Line Graphs

LINE GRAPH

Shows how something changes over time. Line goes up = increase. Line goes down = decrease.

Graphs show HOW THINGS CHANGE

Reading a line graph:

- Line goes UP → the number is increasing
- Line goes DOWN → the number is decreasing
- Line stays FLAT → no change

INSERT IMAGE

Annotated line graph with arrows pointing to rising section labeled 'Increasing', falling section labeled 'Decreasing', and flat section labeled 'No change'

SOCRATIC QUESTIONS

- *What does the direction of the line tell you?*
- *Why is a line graph better than a table for showing change over time?*

Chat Frame: Pick the Evidence

CHAT FRAME

"If a visitor asked 'Has the world's population been going up or down?' — would you show them a table, a bar chart, a line graph, or a photo? Why?"

GUIDE THE DISCUSSION

- **Table?** Could list population numbers, but hard to see the trend.
- **Bar chart?** Good for comparing, but doesn't show change over TIME.
- **Line graph? BEST — shows up-or-down over time!**
- **Photo?** Can't show population numbers at all.

Discovery Reveal: Photographs

PHOTOGRAPH

Shows what a place really looks like — captures real details like color, texture, and shape that drawings and charts can't.

Photos show WHAT IT REALLY LOOKS LIKE

No chart or map can show the actual colors, textures, and shapes of a real place.

INSERT IMAGE

Dramatic photograph of a mountain range (such as the Andes or Rockies) showing snow-capped peaks, green valleys, and blue sky — vivid and detailed

DISCOVERY FACT

The very first photograph ever taken was in 1826 — it showed a view from a window in France. It took about eight hours to take that one picture! Today your phone takes one in a fraction of a second.

SOCRATIC QUESTIONS

- What can a photograph show that no chart, table, or map can?
- Why would a curator put a photo next to a map of the same place?

Chat Frame: Photo vs. Map

CHAT FRAME

"What can you learn about a place from a photo that you couldn't learn from a map or chart?"

POSSIBLE STUDENT RESPONSES

- What the actual colors and textures of a place look like
- How tall or steep mountains really are
- What the weather looks like at that moment
- What animals or plants live there
- What buildings or structures look like up close

The Full Curator Toolkit

Maps

show WHERE

Tables

show WHAT FACTS

Charts

show HOW MUCH

Graphs

show HOW THINGS CHANGE

Photos

show WHAT IT REALLY LOOKS LIKE

OPTI SAYS

“Back in Unit 1, you learned to study sources carefully and ask, ‘What kind of evidence is this?’ You’re using that exact same skill right now — just with charts and graphs instead of artifacts. Your curator training is really paying off!”

Discussion Scaffolding

THINKING MOVE: Sort → Compare → Explain

1

SORT

Sort the evidence types by what they do best.

2

COMPARE

Compare two of them — how are they similar? How are they different?

EXPLAIN

SOCRATIC QUESTIONS

- Which two evidence types are most similar? Most different?
- Could you ever need more than one type to answer a single question?

Guided Practice: I Do

EVIDENCE QUESTION 1

Pick two of the four evidence types (table, bar chart, line graph, photograph). For each one, explain what it does and give an example of a question it can answer.

TEACHER MODEL

Evidence Type 1: Bar Chart

A bar chart uses bars of different heights to compare amounts. Taller bar means more. A bar chart could answer: “Which continent has the most people?”

Evidence Type 2: Photograph

A photograph shows what a place really looks like — the real colors, textures, and shapes. A photograph could answer: “What does the Amazon rainforest actually look like?”

Guided Practice: We Do

EVIDENCE QUESTION 2

Why can't a map tell the whole story by itself? Give one example of something a map can't show that another type of evidence can.

SENTENCE SCAFFOLD

A map can show ____, but it can't show ____.

For that, you'd need a ____.

SAMPLE ANSWER (build together)

"A map can show where a river is, but it can't show exactly how long it is in miles. For that, you'd need a table."

Guided Practice: You Do

EVIDENCE QUESTION 3

How does today's lesson connect to what you learned in Unit 1 about studying sources carefully? What skill are you using in both?

THINKING HINTS

- In Unit 1, what did you learn about sources and evidence?
- What question did you always ask?
- Are you asking a similar question today?
- What's the skill that connects both units?

LOOK FOR ANSWERS LIKE

"In Unit 1, we learned to look at evidence carefully and ask what kind it is. Today we're doing the same thing, but with charts and graphs. The skill is figuring out what type of evidence you're looking at and what it can tell you."

Give students 3–4 minutes to write independently. Then share 2–3 responses.

Error Analysis

STUDENT SAMPLE

“A bar chart is the best evidence for everything because bars are easy to read.”

What's wrong with this answer? Can one type of evidence really be the best for EVERY question?

THE FIX

Each evidence type answers a DIFFERENT kind of question — you have to match the evidence to the question.

- Want to compare amounts? → Bar chart
- Want to see change over time? → Line graph
- Want specific facts? → Table
- Want to see what it really looks like? → Photograph

Tricky One: Curator Task Practice

VISITOR QUESTION

“I want to know whether the world’s biggest cities are getting bigger or smaller over time.”

THINK THROUGH IT

What kind of question is this?

→ It asks about CHANGE OVER TIME.

Which evidence type shows change over time?

→ A LINE GRAPH.

Why not the others?

→ A table gives numbers but not the visual trend.

→ A bar chart compares amounts, not change.

→ A photo can’t show data.

CURATOR TASK PREVIEW

On the assignment, you’ll write 2–3 sentences:

- Which type of evidence would you recommend?
- Why is it better for this question than the other types?

Use what we just practiced to help you write a strong answer.

Self-Check: Exhibit Label

KEY TAKEAWAYS

- Maps show WHERE
- Tables show WHAT FACTS
- Bar charts show HOW MUCH
- Line graphs show HOW THINGS CHANGE
- Photos show WHAT IT REALLY LOOKS LIKE

Match the evidence to the question!

SELF-CHECK QUESTIONS

1. Which type of evidence is best for showing how something changes over time?

A) Table B) Line graph C) Photo

Answer: B — Line graph

2. A curator wants to compare the populations of four continents. Which evidence type would work best?

A) Bar chart B) Photo C) Line graph

Answer: A — Bar chart

3. Why would a curator display a photograph next to a map of the same place?

C) Because a photo shows real details like color, texture, and shape that a map can't

Assignment Preview

EVIDENCE QUESTIONS

1. Pick two of the four evidence types (table, bar chart, line graph, photograph). For each one, explain what it does and give an example of a question it can answer.
2. Why can't a map tell the whole story by itself? Give one example of something a map can't show that another type of evidence can.
3. How does today's lesson connect to what you learned in Unit 1 about studying sources carefully? What skill are you using in both?

CURATOR TASK: EVIDENCE RECOMMENDATION

A visitor walks into the Great Map Room and asks: "I want to know whether the world's biggest cities are getting bigger or smaller over time."

Write 2–3 sentences: Which type of evidence would you recommend? Why is it better for this question than the other types?

CURATOR'S CHRONICLE

In this exhibit, I learned that _____ because _____.

TURN-IN CHECKLIST

☒ Evidence questions ☒ Curator Task ☒ Chronicle ☒ Submit to Canvas

GREAT WORK, MAP CURATOR!

Today you learned to read tables, charts, graphs, and photos — and to choose the right evidence for the right question. That's real curator thinking.

COMING NEXT

Next time, the museum director is going to ask you to prove that you can build a clear map from scratch with all the essential elements — title, compass rose, scale, and legend.